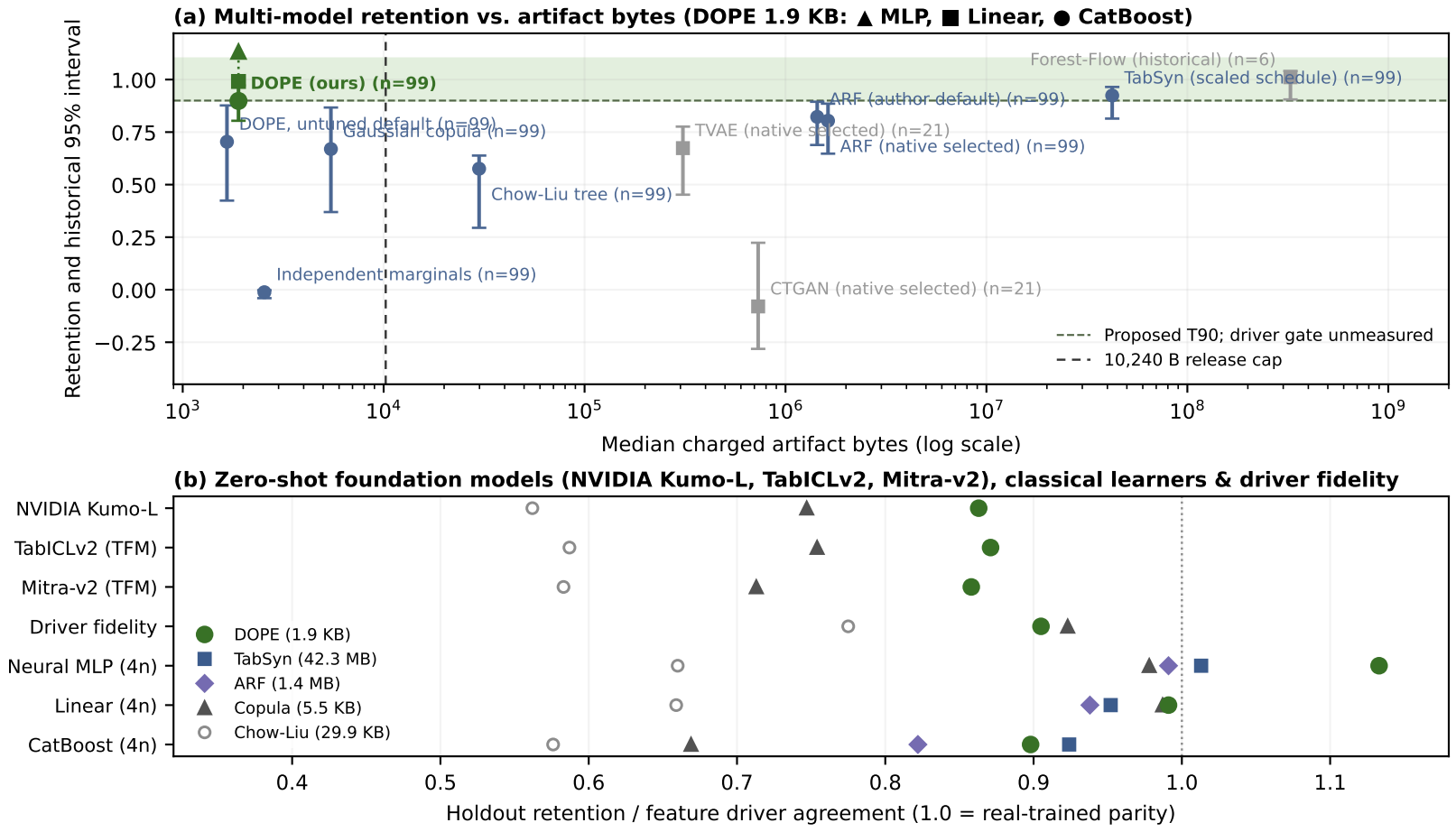
(a) Multi-model retention vs. artifact bytes (DOPE 1.9 KB: ▲ MLP, ■ Linear, ● CatBoost)
Retention and historical 95% interval
1.00
0.75
0.50
0.25
0.00
−0.25
DOPE (ours) (n=99)
DOPE, untuned default (n=99)
Gaussian copula (n=99)
Chow-Liu tree (n=99)
Independent marginals (n=99)
TVAE (native selected) (n=21)
ARF (author-default) (n=99)
ARF (native selected) (n=99)
CTGAN (native selected) (n=21)
Forest-Flow (historical) (n=6)
TabSyn (scaled schedule) (n=99)
Proposed T90; driver gate unmeasured
10,240 B release cap
$10^3$
$10^4$
$10^5$
$10^6$
$10^7$
$10^8$
$10^9$
Median charged artifact bytes (log scale)
(b) Zero-shot foundation models (NVIDIA Kumo-L, TabICLv2, Mitra-v2), classical learners & driver fidelity
NVIDIA Kumo-L
TabICLv2 (TFM)
Mitra-v2 (TFM)
Driver fidelity
Neural MLP (4n)
Linear (4n)
CatBoost (4n)
DOPE (1.9 KB)
TabSyn (42.3 MB)
ARF (1.4 MB)
Copula (5.5 KB)
Chow-Liu (29.9 KB)
0.4
0.5
0.6
0.7
0.8
0.9
1.0
1.1
Holdout retention / feature driver agreement (1.0 = real-trained parity)